POWER INPUT: EXTERNALLY PROTECTED 2S LI-ION | 6.0 ... 8.4 V | NO CHARGER ON CARRIER PCB

WARNING: TURN OFF / DISCONNECT BATTERY POWER BEFORE CONNECTING ESP32 DEVKIT USB-C TO A COMPUTER.

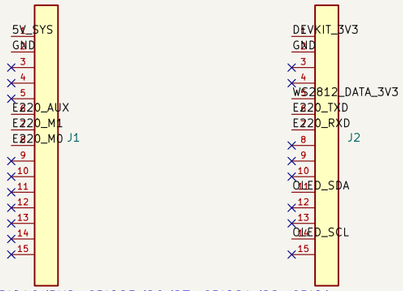
ESP32 DEVKIT, E220 AND OLED ARE USER-INSTALLED MODULES.

Inter-block links use net labels. Local wires show the active power, bypass and interface support networks.

C. REMOVABLE ESP32 DEVKIT

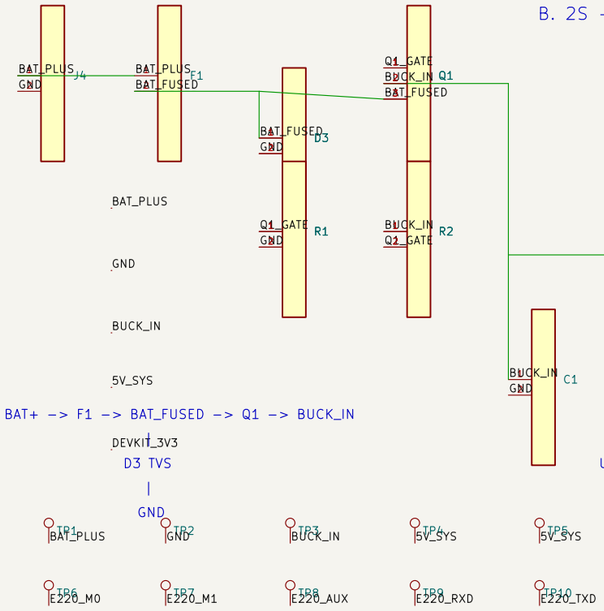
30 PIN / 2x15 | USB-C + CH340C ON MODULE

LEFT HEADER: J1 RIGHT HEADER: J2



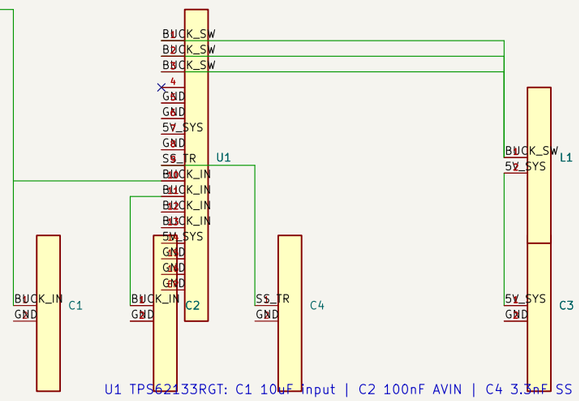
USED: VIN, 3V3, GND | GPIO17/TX2, GPIO16/RX2, GPIO25/26/27, GPIO21/22, GPIO4

A. BATTERY INPUT / PROTECTION



BAT+ -> F1 -> BAT_FUSED -> Q1 -> BUCK_IN

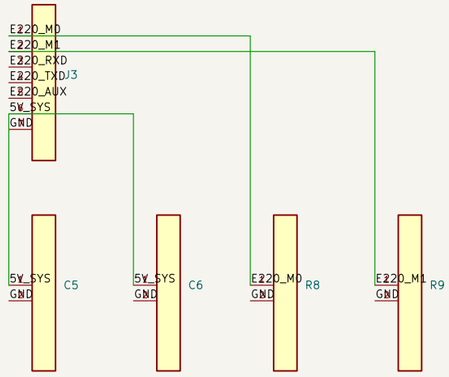
B. 2S -> 5V BUCK



U1 TPS62133RGT: C1 10uF input | C2 100nF AVIN | C4 3.3nF SS | L1 2.2uH | C3 22uF output

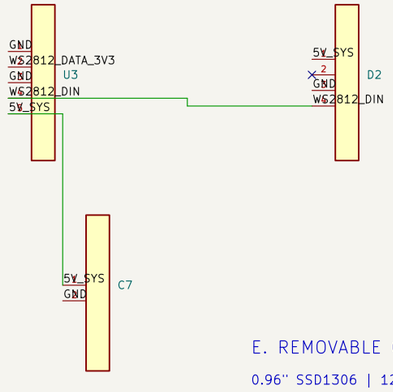
D. E220-T22D UNIVERSAL SOCKET

E220-400T22D / E220-900T22D - user-installed module



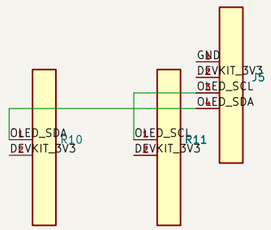
F. WS2812 STATUS LED

GPIO4 -> AHCT buffer -> DIN



E. REMOVABLE OLED

0.96" SSD1306 | 128x64 | I2C | USER INSTALLED



J5: 1 GND | 2 DEVKIT_3V3 | 3 GPIO22/SCL | 4 GPIO21/SDA | R10/R11 4.7k DNP

Sheet: /
File: esp32-e220.kicad_sch

Title: ESP32 E220 LoRa Receiver - Rev.1 protected 2S battery carrier

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 1/1